Exercises (Algebra-1)

October 8, 2026

Notations: Let us denote $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}_n$.

1. Let G be the set of all rotations of the plane about the origin in the plane and $\circ$ the composition of mappings. Thus, prove that $(G, \circ)$ is a group.

2. Consider the regular n-gon (polygon of n equal sides and equal internal angles) inscribed in the unit circle in the plane, so that one of the vertices is $(1, 0)$. Let R_n be the set of all rotations about the origin which maps this regular n-gon into itself. Prove that $(R_n, \circ)$ is a group, where $\circ$ is the composition of mappings. How many elements are there in this group?

3. Let D_n be the set of all rotations and reflections which map the regular n-gon, given in above, into itself. Then prove that $(D_n, \circ)$ is a group, where $\circ$ is the composition of mappings. How many elements D_n has? The group $(D_n, \circ)$ is called the dihedral group of degree n. The elements of D_n are called the symmetries of the regular n-gon.

4. Let n be a positive integer and consider the group $(\mathbb{Z}/n\mathbb{Z}, +_n)$ of integers modulo n. For any $0 < d < n$, prove that $\mathbb{Z}/n\mathbb{Z} = \langle d \rangle$ if and only if d is relatively prime with n.

5. Determine all the subgroups of each of the following:

   (i) $(\mathbb{Z}/24\mathbb{Z}, +_{24})$

   (ii) $(\mathbb{Z}/7\mathbb{Z}, +_7)$

6. For any subgroup H of a group G and $a \in G$, prove that $aHa^{-1} = \{axa^{-1} : x \in H\}$ is also a subgroup of G.

7. List all the elements in each of the following subgroups of the groups mentioned:

   (i) $\langle 7 \rangle$ in $(\mathbb{Z}_{18}, +_{18})$

   (ii) $\langle 5 \rangle$ in $(\mathbb{Z}_{20}, +_{20})$

   (iii) $\langle 3 \rangle$ in $(\mathbb{Z}_{12}, +_{12})$

   (iv) $\langle 3 \rangle$ in $(\mathbb{Z}_{16}, +_{16})$

8. Determine which of the following are morphisms of groups between the given groups:

   (i) Consider the group $(\mathbb{Z}, +)$ and define $f : \mathbb{Z} \to \mathbb{Z}$ by $f(a) = 2a$ for all $a \in \mathbb{Z}$.

   (ii) Consider the groups $(\mathbb{Z}_6, +_6)$ and $(\mathbb{Z}_8, +_8)$ and define $f : \mathbb{Z}_6 \to \mathbb{Z}_8$ by $f(a) = a$ for all $a \in \mathbb{Z}_6$.

   (iii) Consider the groups $(\mathbb{Z}_6, +_6)$ and $(\mathbb{Z}, +)$ and define $f : \mathbb{Z}_6 \to \mathbb{Z}$ by $f(a) = a$ for all $a \in \mathbb{Z}_6$.

   (iv) Define $f : \mathbb{Z}_8 \to \mathbb{Z}_2$ by $f(a) = 0$ if a is even, and 1 if a is odd.

   (v) Define $f : \mathbb{Z}_{15} \to \mathbb{Z}_2$ as in (iv) above.

9. Let $\mathbb{R} \setminus \{0\}$ be the group of nonzero real numbers under multiplication and $G = (\{1, -1\}, \cdot)$. Define $f : \mathbb{R} \to G$ by $f(a) = 1$ if $a > 0$, and -1 if $a < 0$. Then, prove that f is a surjective morphism of groups.

10. Prove that there is no surjective morphism of groups from $(\mathbb{Z}, +)$ onto $(\mathbb{R}, +)$.

11. List all the isomorphisms of the group $(\mathbb{Z}, +)$ onto $(\mathbb{Z}, +)$.

12. Suppose that $g^2 = e$ for all elements g of a group G; prove that G is commutative.

13. Prove that no two of the groups $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ are isomorphic to one another. Can you decide whether $(\mathbb{R}, +)$, $(\mathbb{C}, +)$ are isomorphic to one another?

14. Let G be a group. Prove that the function $G \to G$ defined by $g \mapsto g^{-1}$ is a morphism of groups if and only if G is commutative. Prove that $g \mapsto g^2$ is a morphism of groups if and only if G is commutative.

15. Let G be a commutative group, and let $n > 0$ be an integer. Prove that $\{g^n \mid g \in G\}$ is a subgroup of G. Prove that this is not necessarily the case if G is not commutative.

16. Exhibit an injective morphism of groups from $\mathbb{Z}_8$ into $\mathbb{Z}_{24}$.

17. For any positive integers m and n, obtain a necessary and sufficient condition for having an injective morphism of groups from $\mathbb{Z}_m$ into $\mathbb{Z}_n$.

18. How many morphisms of groups are there from $\mathbb{Z}_7$ into $\mathbb{Z}_{15}$?

19. Let a be an element in a group G and $n \in \mathbb{Z}^+$. Define

$$f : \mathbb{Z}_n \to G \quad \text{by} \quad f(i) = a^i$$

    for $0 \leq i < n$. Obtain a necessary and sufficient condition in terms of a and n for f to be a morphism of groups.

20. Let $n \in \mathbb{Z}^+$ and G be the group of all nth roots of unity under the usual multiplication of complex numbers. Prove that G is isomorphic to $(\mathbb{Z}_n, +_n)$.

21. Let G and G' be groups and $f : G \to G'$ be a mapping. Prove that f is a morphism of groups if and only if

$$f(ab^{-1}) = f(a)f(b)^{-1}$$

for all a and $b \in G$.

22. Prove that a nonempty subset H of a group G is a subgroup of G if and only if $HH^{-1} \subseteq H$. (Here $HH^{-1} = \{hk^{-1} \mid h, k \in H\}$.)